SQL Practice Questions – LEFT, RIGHT, CASE, Subqueries, Execution Order

1. 1. Show the first 3 letters of each customer's first name.
2. 2. Display the last 2 characters of each product's name.
3. 3. Return the first 4 letters of vendor names.
4. 4. Get the last 3 characters of all customer ZIP codes.
5. 5. List vendor owner first names with their first initials using LEFT().
6. 6. Extract the last 5 letters of product descriptions using RIGHT().
7. 7. Display a derived column with the first 2 letters of each market day.
8. 8. Show vendor booth numbers and their last digit.
9. 9. Create a code from the first 2 letters of vendor name and last 2 digits of vendor ID.
10. 10. Return the first and last letter of each product category name.
11. 11. Create a column is_long_name — 1 if customer_first_name is longer than 5 characters, else 0.
12. 12. In the vendor_inventory table, create a column to classify original_price as 'Low', 'Medium', or 'High'.
13. 13. Label booths as 'Premium' if booth_price_level is 'A', 'Standard' if 'B', else 'Basic'.
14. 14. Add a column in customer table: 'Metro' if ZIP > 60000, 'Urban' if between 30000–60000, else 'Rural'.
15. 15. Create a season_label in market_date_info — 'Hot' if max temp > 30, else 'Normal'.
16. 16. Use CASE in customer_purchases to label quantity as 'Bulk' (>10), 'Standard' (>5), or 'Few'.
17. 17. Flag products with a product_size ending in 'kg' as 'Weight-Based' or else 'Unit-Based'.
18. 18. In vendor, show 'Owner Match' if vendor_name contains the vendor_owner_last_name, else 'Independent'.
19. 19. Return a new column market_day_type — 'Weekend' if Saturday/Sunday, else 'Weekday'.
20. 20. Classify ZIP codes as Short (<=5 digits) or Long using CASE.
21. 21. Flag product_name as 'Has Green' if it contains 'Green', else 'Other'.
22. 22. Add a special_flag to market dates where special_notes is not NULL.
23. 23. Return vendor name and assign category using CASE: if vendor_type contains 'Fruit', then 'Fruit Seller', else 'General'.
24. 24. Create a column in booth to show 'Ends in 0' if booth_number ends in 0.
25. 25. Assign customer type using CASE: 'A' for ZIP < 30000, 'B' for ZIP < 60000, else 'C'.
26. 26. Get the list of all customers whose customer_zip appears more than once in the customer table by matching with other customers' ZIPs.
27. 27. Retrieve all vendors from the vendor table who have at least one matching vendor_id present in the vendor_inventory table.

28. 28. List all products from the product table that have been purchased at least once (i.e., their product_id exists in the customer_purchases table).
29. 29. Find all vendors whose vendor_id does not appear in the vendor_booth_assignments table.
30. 30. Return all product IDs from the vendor_inventory table that do not appear in the customer_purchases table.
31. 31. Get customer IDs from the customer_purchases table where the market_date matches any market date in the market_date_info table with market_snow_flag = 1.
32. 32. List all booth_number values from the vendor_booth_assignments table where the market_date falls in a Winter season, as per the market_date_info table.
33. 33. Retrieve customer_id and market_date from the customer_purchases table for all purchases where the vendor_id is 5.
34. 34. List all products where the product_category_id matches any value in the product_category table.
35. 35. Return all market_date entries from the vendor_booth_assignments table that are not listed in the market_date_info table.
36. 36. Find all vendor_id values that exist in the vendor_booth_assignments table but do not exist in the vendor_inventory table.
37. 37. Get the list of customers from the customer table who made purchases (customer_id in customer_purchases) for products that do not exist in the vendor_inventory table.
38. 38. Retrieve all vendors where the vendor_owner_last_name appears as a customer_last_name in the customer table.
39. 39. List all product_id values from the product table where the product_category_id is also present in the vendor_inventory table.
40. 40. Find all customers whose customer_first_name starts with the same letter as any vendor's vendor_name.
41. 41. Write a query that uses WHERE to filter first, then ORDER BY to sort by customer_last_name.
42. 42. Display vendors whose names start with "G", then sort by vendor_type descending.
43. 43. In product, show records with product_qty_type = 'kg', ordered by product_name.
44. 44. Select first 10 rows from customer_purchases, ordered by market_date descending.
45. 45. First filter product_size not NULL, then return only those starting with a digit.
46. 46. Show all vendors where vendor_type is not null, sorted alphabetically by name.
47. 47. In market_date_info, filter market_day = 'Saturday', then select the earliest market_date.
48. 48. Return products with name starting with "T", limit to 5, sorted alphabetically.
49. 49. Show all customers whose ZIP ends in '0' and order by first name.
50. 50. From vendor_inventory, first filter quantity > 5, then order by original_price descending and limit to 5 rows.